

Coding Implementation Guide

Required results, coupled Monte Carlo paths, and reproducible numerical experiments

Core responsibility. Implement an exact-error GBM benchmark and a numerical-reference non-affine study. For each model, produce the required evidence listed before its algorithms. One command must regenerate every table and figure.

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|----------------------------|-----------------------------------|
| 1. Code structure | 2. Configuration |
| 3. Function interfaces | 4. Path coupling |
| 5. GBM results and methods | 6. Non-affine results and methods |
| 7. Statistics | 8. Required tests |

1. Code structure and rules

code/	
config.py	parameters, grids, path counts, seed
random_paths.py	Brownian generation and aggregation
gbm.py	exact, Euler–Maruyama, Milstein
nonaffine.py	non-affine Euler–Maruyama
statistics.py	errors, intervals, slope fitting
experiments.py	experiment pipelines
plotting.py	report figures
run_all.py	regenerates every result
tests/	numerical tests
figures/	generated PDF figures
results/	generated CSV tables

- Use double precision and `numpy.random.Generator` with seed `32104`.
- Generate increments outside solvers and pass arrays explicitly.
- Vectorise over paths; loop only over time.
- Use batches of about 10,000 paths for GBM and 1,000 for the 8,192-step non-affine master grid.
- Save numerical CSV data before plotting.

2. Reproducible configuration

GBM team defaults

$$S_0=100, \quad \mu=0.05, \quad \sigma=0.20, \quad T=1$$

Setting	Default	Reason
Steps	8, 16, 32, 64, 128, 256, 512	Nested convergence grids
Paths	200,000 initially	Increase if weak bias is hidden by sampling error
Distribution	$N=256$	Representative fine grid

The nonzero drift is required because the weak bias of the mean vanishes for $\mu=0$.

Prescribed non-affine benchmark

$$S_0=100, \quad X_0=\log 100, \quad Y_0=0, \quad T=1$$

$$\mu=0.05, \quad \kappa=2, \quad \theta=-0.2, \quad \xi=0.6$$

$$\rho=-0.7, \quad \sigma_{\min}=0.10, \quad \sigma_{\max}=0.50$$

Setting	Default	Use
Coarse steps	32, 64, 128, 256, 512	Strong and weak refinement
References	4,096 and 8,192 steps	Reference-resolution check
Paths	50,000 initially	Increase if payoff intervals are too wide

3. Core function interfaces

```
gbm_exact_terminal(S0, mu, sigma, T, dW) -> S_T
gbm_em_terminal(S0, mu, sigma, h, dW, track_negative=False)
gbm_milstein_terminal(S0, mu, sigma, h, dW, track_negative=False)
aggregate_increments(dB_fine, ratio) -> dB_coarse
correlate_increments(dB1, dB2, rho) -> dW1, dW2
nonaffine_em_terminal(params, h, dW1, dW2) -> X_T, Y_T, S_T
mean_se_ci(samples) -> mean, standard_error, lower, upper
fit_loglog_slope(step_sizes, errors, valid_mask) -> slope, intercept
```

Increment arrays have shape `(number_of_paths, number_of_steps)`. Solvers must never draw their own randomness.

4. Critical path-coupling rules

“Same Monte Carlo path” is precise. For fixed path index m , every matched method and resolution uses the same Brownian realization. Different indices $m=1,\dots,M$ remain independent. Never reuse one path as all Monte Carlo samples.

GBM

- Exact, EM, and Milstein receive the identical fine increment array.
- Every coarse increment is a block sum of fine increments:

$$\Delta W_n^{(h)} = \sum_{j=nR}^{(n+1)R-1} \Delta W_j^{(h_{\text{fine}})}, \quad h=R \ h_{\text{fine}}$$

- The exact terminal value uses the same $W_T = \sum_n \Delta W_n$.
- Fresh increments for each method would measure path variation, not discretisation error.

Non-affine model

Generate one independent pair of 8,192-step master streams $\Delta B_1, \Delta B_2$. Sum each separately for every coarser grid, then construct

$$\Delta W^{(1)} = \Delta B_1$$

$$\Delta W^{(2)} = \rho \Delta B_1 + \sqrt{1-\rho^2} \Delta B_2.$$

All coarse solutions and both references descend from these master streams. Paired weak differences use the same common random numbers.

A common seed is insufficient by itself. Different call orders or array shapes can advance the generator differently. Generate a master array once, aggregate explicitly, and pass it into the solvers.

5. GBM: required results and implementation

5.1 Results to produce

Result	Required evidence	Files
Strong convergence	EM and Milstein exact errors, 95% intervals, fitted range, slopes near 1/2 and 1	gbm_strong.csv gbm_strong_convergence.pdf
Weak convergence	Signed mean bias, intervals, analytic bias, resolved order-one fit	gbm_weak.csv gbm_weak_convergence.pdf
Distribution	Numerical histogram, exact density, 5/25/50/75/95% quantiles	gbm_distribution.csv gbm_distribution.pdf
Moments	Numerical and exact mean and variance, with M and h	gbm_moments.csv
Positivity (optional)	Any-negative-path and negative-terminal rates kept separate	Table or figure

5.2 Schemes

$$S_T = S_0 \exp[(\mu - \sigma^2/2)T + \sigma \sum_n \Delta W_n]$$

$$S_{n+1}^{\text{EM}} = S_n^{\text{EM}} (1 + \mu h + \sigma \Delta W_n)$$

$$S_{n+1}^{\text{Mil}} = S_n^{\text{Mil}} [1 + \mu h + \sigma \Delta W_n + \frac{1}{2} \sigma^2 ((\Delta W_n)^2 - h)]$$

Do not apply EM to the GBM log-price for convergence; that update is the exact transition.

5.3 Strong, weak, and distribution statistics

$$D_m(h)=|S_N^{(m)}-S_T^{(m)}|$$

$$B_m(h)=S_N^{(m)}-S_T^{(m)}$$

$$\delta(h)=S_0[(1+\mu h)^{T/h}-e^{\mu T}]$$

Fit strong slopes only on the resolved fine-grid range. For weak error, inspect the signed interval before taking an absolute value. Fit only if the interval excludes zero and its half-width is below half the estimated bias.

$$E[S_T]=S_0e^{\mu T}, \quad \text{Var}(S_T)=S_0^2e^{2\mu T}(e^{\sigma^2 T}-1)$$

At $N=256$, retain numerical terminal samples for the histogram, quantiles, mean, and variance. Exact samples alone do not test EM or Milstein.

6. Non-affine model: required results and implementation

6.1 Results to produce

Result	Required evidence	Files
Increment diagnostics	Means and variances versus o and h ; correlation versus ρ	Validation table
Strong refinement	Coupled errors, 95% intervals, empirical slope, 4,096 versus 8,192 reference check	<code>nonaffine_strong.csv</code> <code>nonaffine_strong_convergence.pdf</code>
Weak payoff refinement	Signed paired differences, intervals, resolved slope, both references	<code>nonaffine_weak.csv</code> <code>nonaffine_weak_convergence.pdf</code>
Path validation	Counts of non-finite X , Y , S ; finite $S=e^{X>0}$	Validation summary

6.2 Euler–Maruyama

$$g_n=\sigma_{\min}+(\sigma_{\max}-\sigma_{\min})/(1+e^{-Y_n})$$

$$X_{n+1}=X_n+(\mu-g_n^2/2)h+g_n\Delta W_n^{(1)}$$

$$Y_{n+1}=Y_n+\kappa(\theta-Y_n)h+\xi\sqrt{(1+Y_n^2)}\Delta W_n^{(2)}$$

$$S_T=e^{X_T}$$

Fail clearly on non-finite values; never clip or silently drop paths. Do not copy the scalar GBM Milstein correction into this two-noise model.

6.3 Strong and weak refinement

$$A_m(h)=|S_T^{(h,m)}-S_T^{(h_{\text{ref}},m)}|$$

$$\varphi(s)=\max(s-100,0), \quad P_m(h)=\varphi(S_T^{(h,m)})-\varphi(S_T^{(h_{\text{ref}},m)})$$

Report means, standard errors, and intervals. Repeat with both references. Call the strong slope empirical, and do not claim a slope when sampling uncertainty is comparable with the estimated difference.

7. Statistical and plotting rules

$$\bar{U}=(1/M)\Sigma_m U_m, \quad s_U^2=[1/(M-1)]\Sigma_m (U_m-\bar{U})^2$$

$$95\% \text{ interval: } \bar{U} \pm 1.96 s_U / \sqrt{M}$$

- Use the sample standard deviation with one degree-of-freedom correction.
- Show axes, path count, intervals, and fitted range.
- Use log–log axes for convergence.
- Do not fit below the Monte Carlo noise floor.
- Record seed, parameters, path count, and fitted points in CSV.

8. Required tests

- ☐ The seed reproduces all CSV results.
- ☐ Every coarse increment equals its fine block sum to round-off.
- ☐ For each path, exact GBM, EM, and Milstein receive identical increments.
- ☐ Every non-affine grid descends from the same 8,192-step master pair.
- ☐ Increment means, variances, and correlation agree with σ , h , and ρ within uncertainty.
- ☐ The exact GBM terminal formula agrees with repeated exact steps.
- ☐ Resolved strong slopes approach $\frac{1}{2}$ and 1.
- ☐ Simulated weak bias agrees with the analytic bias within its interval.
- ☐ GBM moments and quantiles agree with exact values within uncertainty.
- ☐ Every non-affine terminal state is finite and every price is positive.
- ☐ The 4,096 and 8,192 reference conclusions agree.
- ☐ `python code/run_all.py` recreates every table and figure.

9. Definition of done

Complete means: all required result tables and figures exist; solvers receive externally generated, correctly coupled increments; every estimate includes uncertainty; all checks pass; and `run_all.py` reproduces the complete output. Call GBM an *exact-error study* and the non-affine work a *numerical-reference refinement study*.